

CITS4404 Project — Building AI Trading Bots: Nature-Inspired Optimisation Algorithms Implementation and Experimentation, and Evaluation

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I. ALGORITHMS INVESTIGATED

Six optimisation algorithms were evaluated across all experimental configurations, spanning four structural categories: swarm-based, physics-inspired, ecosystem-inspired, and classical local search.

Particle Swarm Optimization, originally proposed by Kennedy and Eberhart [1] and extended with an inertia weight by Shi and Eberhart [2], is a swarm method in which each particle maintains both a personal best position and a shared global best. The inertia weight w governs the balance between exploration and exploitation through the velocity update rule. Atomic Orbital Search (AOS) [3] is a physics-inspired method detailed in Section III; it employs shell-based clustering around a nucleus (global best) as its primary exploration mechanism, without per-agent personal memory. African Buffalo Optimization (ABO) [4] encodes two herd vocalisations — an exploratory foraging call and an exploitative regrouping call — and like PSO retains both personal and global best positions. Manta Ray Foraging Optimization (MRFO) [5] models three foraging behaviours (chain, cyclone, and somersault foraging) but maintains only a global best, with no per-agent memory. Symbiotic Organisms Search (SOS) [6] simulates mutualistic, commensal, and parasitic ecological interactions, applying all three phases to every organism per iteration; this yields three fitness evaluations per organism per iteration and requires no algorithm-specific hyperparameters. Generalized Pattern Search (GPS) [7] is a classical single-state local optimiser that evaluates a fixed pattern of directions from the current best solution, with no population and no historical memory beyond the current position; it serves as a deliberate baseline to quantify the value of population-based diversity.

Table I summarises the distinguishing characteristics of each algorithm.

TABLE I. Summary of the six optimisation algorithms evaluated. Memory column describes which historical positions are retained per agent. Eval/iter denotes fitness evaluations per agent per iteration.

Algorithm	Year	Category	Population-based	Memory	Eval/iter
AOS	2021	Physics	Yes	Global (nucleus)	1
PSO	1998	Swarm	Yes	Personal + Global	1
MRFO	2020	Swarm	Yes	Global only	1
SOS	2014	Ecosystem	Yes	Global only	3
ABO	2015	Swarm	Yes	Personal + Global	1
GPS	—	Local search	No	Current best only	1

II. BOT DESIGN AND PARAMETRISATION

A. Trading Signal and Fitness

Each strategy produces a short-window (low-period) and a long-window (high-period) smoothed price series. The trading signal detects crossover events in their difference using a two-point weighted kernel:

$$d_t = \text{Series}_{\text{short},t} - \text{Series}_{\text{long},t} \quad , \quad C_t = \frac{1}{2}(\text{sign}(d_t) - \text{sign}(d_{t-1})) \quad (1)$$

$$S = \begin{cases} \text{buy} & \text{if } C_t = -1 \text{ (short crosses below long)} \\ \text{sell} & \text{if } C_t = +1 \text{ (short crosses above long)} \end{cases} \quad (2)$$

where d_t is the difference of the two smoothed series at time t , $C_t \in \{-1, 0, +1\}$ is the crossover indicator, and S is the resulting trade action. The signal is non-zero only at the moment of a crossover; it is zero on all other timesteps.

The fitness function measures final USD holdings after simulating trades over the full training price series:

$$\text{Fitness}(X) = \text{USD}_{\text{final}} \quad (3)$$

where X is the candidate parameter vector, $\text{USD}_{\text{final}}$ is the cash held at the end of the simulation, and the starting balance is $\text{USD}_{\text{initial}} = \1000 . A transaction fee of 3% is applied to each trade. The simulation is a single continuous pass over the training data with no mid-run resets.

One acknowledged limitation is that the fitness function imposes no penalty for non-trading: if no crossover ever occurs, $C_t = 0$ for all t and the bot holds cash throughout, returning $\text{Fitness}(X) = \text{USD}_{\text{initial}}$. This is a valid but uninformative fixed point. The degenerate optimum was observed empirically — MRFO converged to parameter regions with no crossover signal under certain configurations. A risk-adjusted metric such as the Sharpe ratio would penalise this behaviour but was not adopted here; the limitation is documented rather than silently corrected.

B. Strategy Configurations

Five strategy configurations define the hypothesis space presented to each optimiser.

Strategy 1 — Double SMA crossover ($d = 2$). Both the short window $x_0 \in [2, 50]$ and the long window $x_1 \in [51, 200]$ are optimised. The non-overlapping constraint (short < long) is guaranteed by construction through the parameter bounds, with no runtime enforcement required. The SMA assigns uniform weight to all prices within the window, making it insensitive to the ordering of recent versus older prices.

Strategy 2 — Double LMA crossover ($d = 2$). The parameter structure is identical to Strategy 1 ($x_0 \in [2, 50]$, $x_1 \in [51, 200]$). The LMA assigns linearly decaying weights so that more recent prices contribute more than older ones, providing momentum sensitivity that the uniform SMA cannot capture. The optimiser thus faces the same two-dimensional search space as Strategy 1 but with a smoother series that responds differently to regime changes.

Strategy 3 — EMA with shared α ($d = 3$). A short span, a long span, and a single shared smoothing parameter $\alpha \in (0, 1)$ are optimised jointly. Sharing one α value across both EMAs constrains both series to decay at identical rates; the frequency contrast that drives the crossover signal therefore depends entirely on the span difference rather than on independent decay dynamics. This effectively collapses the degree of freedom to approximately 2.5 effective dimensions in practice.

Strategy 4 — EMA with independent α ($d = 4$). Short span, long span, α_{short} , and α_{long} are each optimised independently, along with additional EMA window parameters. Decoupling the smoothing parameters gives the short-term EMA its own reactivity and the long-term EMA its own stability, restoring the frequency contrast that shared α eliminates. The expanded search space ($d = 4$ versus $d = 3$) carries a higher exploration cost but provides genuinely greater expressiveness: the two EMAs can now operate at qualitatively different timescales simultaneously. Empirically, this configuration produced the highest Kruskal-Wallis H -statistics across all strategy conditions, suggesting that the larger search space amplifies inter-algorithm performance differences rather than masking them.

Strategy 5 — Weighted MA combination ($d = 14$). SMA, LMA, and EMA signals are combined as a weighted sum with learned weights w_1, w_2, w_3 . All three weights are treated as free optimisation dimensions and normalised at evaluation time (each w_i divided by $w_1 + w_2 + w_3$), yielding a 14-dimensional parameter vector that includes window parameters for each MA type plus EMA spans and smoothing factors. Deriving $w_3 = 1 - w_1 - w_2$, as the project brief implies, would create asymmetric optimisation pressure because w_3 is never directly searched; retaining all three as free variables ensures each weight receives equal search effort. The 14-dimensional space is the largest tested, and the curse of dimensionality is a genuine concern when each optimiser is limited to 50 runs.

C. Justified Deviations from Project Brief

TABLE II. Summary of implementation decisions that deviate from the project brief, with justification for each.

Deviation	Brief approach	Our implementation	Justification
EMA normalisation	Optional	Enabled by default	Unnormalised kernels produce span-dependent amplitude distortion, making fitness values non-comparable across parameter configurations
$d = 14$ weight derivation	w_3 derived as $1 - w_1 - w_2$	All three weights free; normalised at eval time	Derived w_3 receives no direct search pressure, creating asymmetric optimisation bias
n_{shells} assignment	Fixed shell count	$\min(n_{\text{shells}}, \text{remaining})$	Prevents degenerate empty shell states in high-dimensional runs where population may be smaller than the configured shell count
EMA α across spans	Shared α (Strategy 3)	Independent α_{short} , α_{long} (Strategy 4)	Shared α constrains both series to identical decay rates, eliminating the frequency contrast that makes the crossover signal informative

III. ALGORITHM SELECTION

All six algorithms described in Section I were evaluated. This section explains the rationale behind each inclusion and details the primary algorithm, AOS.

GPS as a deliberate baseline. The central empirical question is whether population-based global search is necessary on a moving-average (MA) parameter fitness landscape, or whether a single-state local optimiser is sufficient. GPS [7] has no population, no escape mechanism from local optima, and no memory of previously evaluated positions beyond the current best. Including it provides a direct lower bound: if GPS performs competitively, population diversity contributes little; if GPS underperforms consistently, the landscape demands it.

PSO as a population-based benchmark. PSO [2] is one of the most extensively studied metaheuristics and provides a reliable population-based performance reference. It is also structurally related to AOS — both use a global nucleus/best to guide the population — making a paired comparison directly informative about the contribution of personal best memory. If AOS and PSO are statistically indistinguishable, per-particle historical memory may or may not add marginal benefit in this problem domain.

AOS as the physics-based algorithm compared to the others being biology-based. The MA parameter fitness landscape is expected to be multimodal: distinct parameter combinations can produce similar short-term profits through qualitatively different mechanisms, creating multiple local peaks of comparable height. AOS [3] was selected because its shell-based population structure explicitly supports simultaneous exploitation of multiple landscape regions. Candidate solutions (electrons) are assigned to concentric shells around the nucleus (current global best) according to their fitness rank; electrons in inner shells exploit promising regions while outer-shell electrons conduct broader exploration. Transitions between shells are governed by a quantum-mechanical photon absorption and emission analogy: an electron absorbs energy (moves to an outer shell, exploration) when its transition energy falls below the shell's binding energy, and emits energy (moves to an inner shell, exploitation) otherwise. This structured push-pull between shells provides more persistent population spread than PSO's convergent swarm behaviour, which collapses toward the global best over time. AOS was published in 2021 and has not, to our knowledge, been applied to financial parameter optimisation in the literature.

The pseudocode below summarises the core AOS iteration:

```

Initialise N electrons randomly in the search space
Set nucleus := position of the best electron (highest fitness)
Assign electrons to K shells by fitness rank (binding energy)

FOR each iteration t = 1, 2, ..., T:
  FOR each shell k = 1, ..., K:
    Compute shell binding energy E_k
    FOR each electron i in shell k:
      Sample transition energy E
      IF E < E_k (electron absorbs energy):
        Move to outer shell; apply exploration step
      ELSE (electron emits energy):
        Move to inner shell; apply exploitation step

```

```

    Update position via quantum transition equation
    Evaluate fitness of new position
    Apply LE perturbation: randomly displace the
        lowest-energy electron to escape local optima
    Update nucleus if any electron improves best fitness

RETURN nucleus position as best solution found

```

LE degradation — an observed weakness. The Lowest Energy (LE) perturbation step is designed to prevent stagnation by randomly displacing the electron with the worst fitness. In practice, the LE designation is not stable: as other electrons improve their fitness across iterations, the electron currently labelled LE may no longer occupy the position most in need of escape relative to the updated nucleus. The original paper [3] does not address this drift. This was observed empirically across runs and is logged per iteration; it is documented here rather than patched, as any correction would constitute a modification to the published algorithm.

Metaparameter choices. Population size was scaled with dimensionality — larger populations were used for the $d = 14$ weighted strategy to partially offset the curse of dimensionality. The shell count n_{shells} is implemented as an upper cap rather than a fixed assignment: each iteration uses $\min(n_{\text{shells}}, |\text{remaining population}|)$ shells to prevent degenerate empty-shell states that can arise in high-dimensional runs where effective population diversity is reduced. The iteration budget was fixed uniformly across all six algorithms to ensure a controlled comparison of per-iteration effectiveness.

IV. EXPERIMENTS AND EVALUATION

A. Dataset

Daily BTC/USD OHLCV data sourced from Kaggle spanning 2014 to 2022 was used. The training set covers the pre-2020 period (approximately 2014–2019) and the test set covers 2020–2022. The split was not chosen arbitrarily: post-2020 data encompasses the COVID-19 market crash of March 2020, the 2021 bull run, and the 2022 bear market, constituting three qualitatively distinct market regimes. This is an intentional out-of-distribution stress test rather than a random holdout. Its purpose is to measure whether parameter configurations optimised on one regime generalise when the underlying market dynamics shift substantially — a more meaningful evaluation of strategy robustness than a matched-regime random split would provide.

B. Experimental Design

The full factorial design comprised 6 algorithms $\times$ 5 strategies $\times$ 2 initialisation modes $\times$ 50 runs = 3000 total trials. Two initialisation modes were tested: fixed initialisation, in which all runs begin from a deterministic seed position, and random initialisation, in which each run begins from a position drawn uniformly at random from the search space. The paired comparison of fixed versus random initialisation tests sensitivity to starting conditions. Strong dependence on initialisation would indicate algorithmic fragility, with convergence driven by the seed rather than by the search mechanism. A run count of 50 was chosen as a practical compromise between statistical power and the total computational cost of 3000 trials; it is sufficient to apply non-parametric distributional tests but modest relative to what would be preferred for high-variance optimisers.

C. Statistical Analysis

The Kruskal-Wallis H -test was used in preference to one-way ANOVA because the structural properties of optimiser output systematically violate the normality assumption required by ANOVA. Optimisers can produce runs that converge to the degenerate no-trade optimum (fitness near \$1000, the starting balance), occasional jackpot runs that far exceed the typical distribution, and heavy-tailed spread in between. These outliers are not noise — they reflect real behavioural modes of the algorithms — and would distort ANOVA F -statistics. Kruskal-Wallis operates on ranks and is robust to such distributional shapes.

For each of the ten strategy $\times$ initialisation mode combinations, the Kruskal-Wallis test was applied with algorithm as the factor variable, holding strategy and initialisation constant. This isolates the algorithm effect from strategy-level confounds: testing across all strategies simultaneously would conflate the variance introduced by different fitness

landscape shapes with the variance attributable to algorithmic differences. Pairwise post-hoc comparisons used Dunn’s test with Bonferroni correction. A significance threshold of $\alpha = 0.05$ was applied throughout.

V. RESULTS AND ANALYSIS

A. Training Results



FIG. 1. Dunn’s post-hoc pairwise p -values (Bonferroni-corrected) for all strategy $\times$ initialisation combinations — training fitness. Red indicates significant difference ($p < 0.05$); green indicates no significant difference.

The Kruskal-Wallis test was significant across all ten strategy $\times$ initialisation combinations ($p < 10^{-20}$ in every case) as per figure III. Algorithm choice has a universally significant effect on training fitness. The `ema_independent` strategy ($d = 4$) consistently yielded the highest H -statistics, suggesting that decoupling the two EMA smoothing parameters amplifies inter-algorithm performance differences more than further increasing dimensionality (`weighted`

TABLE III. Kruskal–Wallis H -test results comparing performance across optimization algorithms for each strategy and initialization mode combination. All ten configurations show statistically significant differences ($p < 0.001$), indicating that algorithm choice meaningfully affects outcomes regardless of strategy or initialization mode.

Strategy	Start mode	H -statistic	p -value
sma	fixed	119.07	4.95×10^{-24}
sma	random	121.15	1.79×10^{-24}
lma	fixed	114.62	4.31×10^{-23}
lma	random	106.19	2.62×10^{-21}
ema_shared	fixed	114.93	3.72×10^{-23}
ema_shared	random	114.24	5.21×10^{-23}
ema_independent	fixed	151.73	5.71×10^{-31}
ema_independent	random	144.01	2.51×10^{-29}
weighted	fixed	120.69	2.24×10^{-24}
weighted	random	120.88	2.05×10^{-24}

at $d = 14$ produced lower H -statistics) — algorithms with stronger exploration mechanisms appear to extract more from the qualitative change in expressiveness than from raw degrees of freedom.

Four patterns emerge from Dunn’s pairwise post-hoc comparisons, illustrated in Figure 1. First, GPS is significantly worse than every other algorithm in 8/10 training conditions; the exceptions are on **ema_independent**, where MRFO collapses to the no-trade fixed point (\$1000, the starting balance) in $\sim 20\%$ of runs and matches GPS’s lower bound. This confirms that single-state local search struggles to navigate the multimodal MA fitness landscape: without population diversity, GPS converges to the nearest local optimum from its starting position. Second, PSO, SOS, and AOS produced training-fitness distributions that were not statistically distinguishable under Bonferroni-corrected Dunn’s post-hoc comparisons in the majority of strategy $\times$ initialisation conditions (non-significance under Bonferroni is not formal equivalence; we report what the test shows). Despite AOS lacking per-particle personal memory, its shell-based exploration achieves comparable training fitness to PSO, suggesting personal-best tracking contributes little marginal benefit when global-best guidance is combined with structured shell diversity. Third, SOS exhibited unusually low within-cell variance on several conditions. Most notably standard deviation of = \$17 across 30 random-init runs on **weighted** and identical fitness on all 30 **sma** and **lma** runs, consistent with its three-phase update collapsing population diversity within a small number of iterations. SOS also requires three fitness evaluations per organism per iteration, so if compute is taken into account for comparison it would likely disadvantage SOS relative to PSO. Fourth, MRFO and ABO occupy intermediate positions that vary by strategy, with no consistent dominance pattern relative to the PSO-AOS-SOS cluster.

B. Test Results — Performance Against the Passive Baseline

TABLE IV. Runs exceeding the buy-and-hold baseline (\$5,660) under strict $>$ comparison. Counts out of 30 runs per cell; 1,800 trials total. One MRFO run on **ema_shared** (fixed init) converged to exactly \$5,660.26, matching buy-and-hold to six decimal places; it is excluded from the strict-inequality counts but discussed below.

	ema_ind			ema_shared			lma			sma			weighted		
algo	fix	rnd	fix	rnd	fix	rnd	fix	rnd	fix	rnd	fix	rnd	fix	rnd	fix
abo	0	0	1		2	0	0	0	0	0	0	1			3
aos	3	1	0		2	0	0	0	0	0	0	0			1
gps	1	0	0		0	0	0	0	0	0	0	0			0
mrfo	0	0	0		1	0	0	0	0	0	0	0			2
psa	0	0	2		0	0	0	0	0	0	0	0			1
sos	0	0	2		2	0	0	0	0	0	0	0			0
total	4	1	5		7	0	0	0	0	0	0	1			7

A passive buy-and-hold strategy over the 2020–2022 test period, with 3% broker fees, produced \$5,660 from a \$1,000 starting balance. Of the 1,800 optimised bots, only 25 (1.38%) strictly exceeded this baseline; no algorithm $\times$ strategy median exceeded it under any condition (Figure 2). Win rates were heavily stratified by strategy (Table IV):

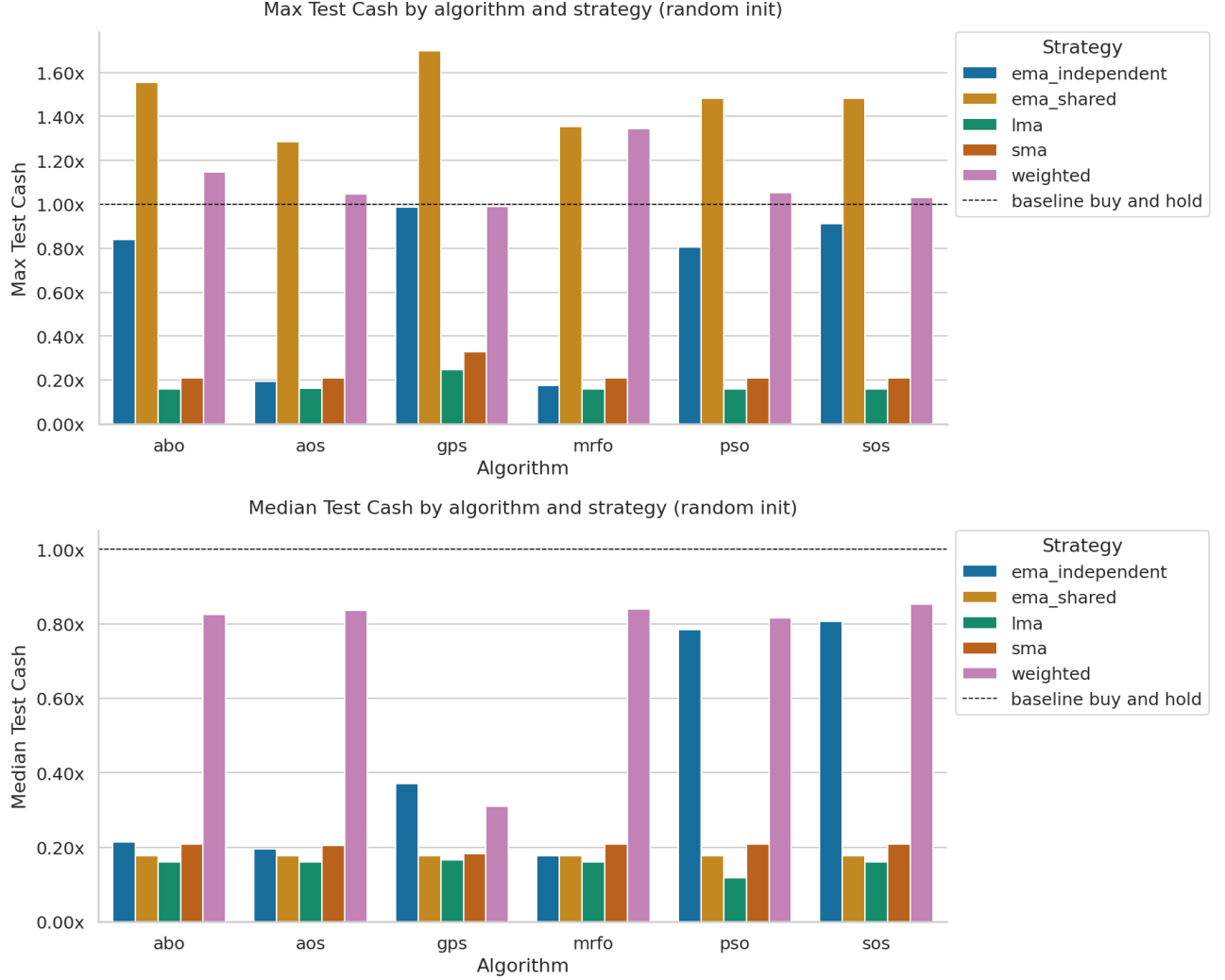


FIG. 2. Median test cash by algorithm and strategy, normalised to the buy-and-hold baseline (\$5,660 from \$1,000 initial under matched 3% entry and exit fees), for fixed (top) and random (bottom) initialisation. No algorithm $\times$ strategy median exceeds the baseline (1.00 $\times$ line) under any condition.

sma and **lma** (both $d = 2$) yielded zero wins across 720 combined runs; **ema_independent** ($d = 4$) produced 5/360 (1.38%); **weighted** ($d = 14$) produced 8/360 (2.22%); **ema_shared** ($d = 3$) produced 12/360 (3.33%).

Using fixed run as reference, according to table V., variance within winning strategies differed substantially. **weighted**-strategy bots converged with median around \$4,600 to \$4,730 (for example PSO $\sigma = \$460$ — , $\mu = \$4,813$). In contrast, **ema_shared** winners came from highly dispersed distributions (PSO $\sigma = \$2,176$ — $\mu = \$2,468$), consistent with random discovery rather than reliable optimisation: a single PSO run on this strategy might produce \$292 (losing money) or maximum single instance of \$10,074, with little predictability from training fitness.

Referring to table VI two MRFO run on **ema_shared** (fixed initialisation, run 14 and random initialisation, run 1) converged to a test fitness of exactly \$5,660.26 matching the buy-and-hold baseline to six decimal places but got excluded. This is not coincidence: the optimiser identified an MA-crossover parameter set that executes precisely one buy at the start of the test period and one sell at the end, replicating passive holding under the matched fee structure.

C. Test Results — Statistical Differentiation

On test data, the significant pairwise differences observed during training largely disappear (Figure 4). The most extreme case is **ema_shared**: under both fixed and random initialisation, every pairwise p -value equals 1.000. All six algorithms produce statistically indistinguishable test fitness distributions. The algorithms that were most differenti-

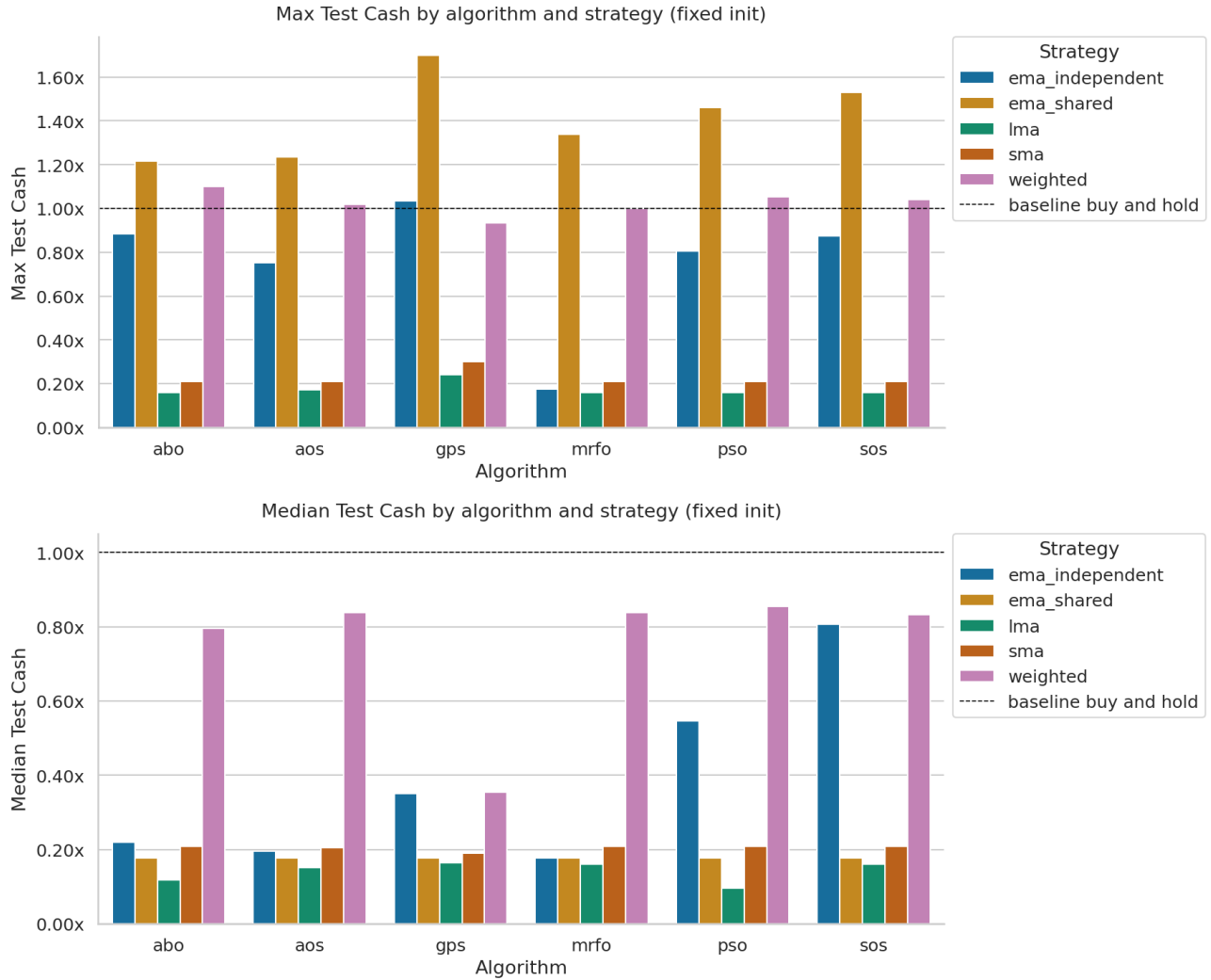


FIG. 3. Maximum test cash by algorithm and strategy, normalised to the buy-and-hold baseline, for fixed (top) and random (bottom) initialisation. Bars above $1.00\times$ indicate at least one run in 30 exceeded passive holding; concentration of such bars in **ema_shared** reflects its long-tailed jackpot distribution rather than reliable optimisation.

ated during training, particularly the PSO-SOS-AOS cluster versus GPS converge to similar test performance across strategies. The exception is **lma** fixed initialisation, where GPS achieves the highest test median (\$948) of all six algorithms. Inspecting GPS, the test median (\$948 **lma** and \$1056 **sma**) is actually higher than train median (\$445 **lma** and \$572 **sma**) this might indicate either (i) coincidence from fixed point initialization or (ii) the greedy local search nature incidentally helped GPS to have better generalization.

D. Why Training Fitness Did Not Predict Test Fitness

Train-test rank correlation (Spearman’s ρ on per-run fitness pairs, both initialisation modes combined) reveals a structural pattern (Figure 5). **sma** ($d = 2$) shows $\rho = 0.67$: if an optimiser ranks high on training, it ranks high on test. **ema_independent** ($d = 4$) shows $\rho = 0.48$, indicating moderate but reliable transfer. **lma** ($d = 2$) follows at $\rho = 0.31$. **weighted** ($d = 14$) is weak and only marginally significant ($\rho = 0.12$, $p = 0.026$). **ema_shared** ($d = 3$) shows no detectable transfer ($\rho = 0.07$, $p = 0.18$, n.s.).

The relationship is not strictly monotonic in dimensionality: **ema_shared** ($d = 3$) has worse transfer than **weighted** ($d = 14$), and the difference between **ema_shared** and **ema_independent** is a single additional parameter, decoupling the two EMAs’ smoothing rates but produces a sevenfold change in ρ . This indicates that the *nature* of the parameterisation matters as much as its size: a tightly-coupled $d = 3$ space can produce less generalisable optima than a

TABLE V. Statistical summary of performance metrics across combinations of algorithms and strategy configurations under the **fixed** start mode. Values represent metrics across 30 runs per cell.

Algo	Strategy	Count	Mean	Std	Var	Min	Median	Max
abo	ema_independent	30	1,482.45	994.69	989,410.42	858.15	1,202.42	4,360.90
	ema_shared	30	1,815.81	1,391.74	1,936,939.29	633.33	1,157.60	6,138.88
	lma	30	683.14	187.41	35,120.68	487.74	671.51	907.98
	sma	30	1,172.25	12.48	155.63	1,153.51	1,180.28	1,180.28
	weighted	30	4,451.70	817.09	667,640.83	1,130.40	4,651.61	5,691.01
aos	ema_independent	30	4,271.85	1,401.70	1,964,770.50	977.93	4,425.39	7,478.46
	ema_shared	30	1,894.03	1,269.02	1,610,421.47	603.28	1,137.12	4,377.15
	lma	30	794.10	176.29	31,077.99	485.41	907.98	921.28
	sma	30	1,172.25	12.48	155.63	1,153.51	1,180.28	1,180.28
	weighted	30	4,465.42	830.24	689,298.14	1,000.00	4,676.32	5,500.04
gps	ema_independent	30	2,328.67	1,619.27	2,262,043.67	509.80	1,588.76	5,684.32
	ema_shared	30	1,831.89	1,465.63	2,148,063.73	462.49	1,000.00	4,584.60
	lma	30	879.14	221.14	48,901.80	378.97	942.98	1,228.54
	sma	30	1,154.99	216.37	46,814.35	725.76	1,104.15	1,740.13
	weighted	30	2,904.46	1,852.43	3,431,485.27	370.96	3,688.60	5,093.26
mrfo	ema_independent	30	1,000.00	0.00	0.00	1,000.00	1,000.00	1,000.00
	ema_shared	30	1,606.10	1,366.90	1,868,412.38	484.18	1,000.00	5,660.26
	lma	30	749.93	200.35	40,140.58	487.74	907.98	907.98
	sma	30	1,180.28	0.00	0.00	1,180.28	1,180.28	1,180.28
	weighted	30	4,329.08	1,145.65	1,312,524.63	1,381.96	4,694.22	5,644.11
pso	ema_independent	30	3,219.79	1,460.65	2,133,495.71	1,201.23	4,296.97	4,565.80
	ema_shared	30	2,468.14	2,176.03	4,735,114.23	824.62	1,000.00	10,074.64
	lma	30	587.25	135.88	18,463.77	487.74	487.74	907.98
	sma	30	1,129.06	142.71	20,365.10	617.37	1,180.28	1,180.28
	weighted	30	4,813.35	460.04	211,639.97	3,637.12	4,729.43	5,564.79
sos	ema_independent	30	3,805.27	1,285.29	1,651,977.33	1,206.81	4,296.97	4,834.21
	ema_shared	30	2,237.57	1,698.33	2,884,337.13	617.00	1,862.03	7,142.09
	lma	30	907.98	0.00	0.00	907.98	907.98	907.98
	sma	30	1,180.28	0.00	0.00	1,180.28	1,180.28	1,180.28
	weighted	30	4,580.57	477.97	228,457.58	2,050.89	4,660.90	4,694.26

decoupled $d = 4$ space or even a $d = 14$ weighted combination.

Ironically, the two strategies with the most test-set wins (**ema_shared**: 13; **weighted**: 8) are also the two with the lowest train-test correlations. Their wins represent stochastic discovery, not reliable optimisation: a reading of $\rho = 0.07$ should not expect any optimiser’s training performance to inform its test outcome on **ema_shared**. The strategies where training rank *does* predict test rank (**sma**, **ema_independent**, **lma**) are predictably mediocre on test meaning their best-training algorithm is also their best-test algorithm, but neither beats the baseline. This is the descriptiveness-versus-search-space trade-off realised empirically: higher expressiveness produced higher training fitness (\$100,000+ medians on **ema_shared**) but transferred no useful information to held-out test data.

E. Which Algorithm Performs Best

Across fixed-initialisation test results as per table VII, no algorithm demonstrated consistent dominance. PSO and SOS produced the strongest combination of median performance and stability on **weighted** (PSO: median = \$4,729, CV = 0.095; SOS: median = \$4,661, CV = 0.104), while AOS and SOS led on **ema_independent** (AOS median = \$4,425; SOS median = \$4,297). Apparent stability on **sma** and **lma** (MRFO, SOS: CV = 0.000) reflects convergence to degenerate fixed points rather than reliable optimisation — zero variance indicates collapse, not robustness. This strategy-dependence of relative algorithm performance is consistent with the No Free Lunch theorem [8]: no algorithm dominates uniformly across fitness landscapes, and the landscape structure — determined by strategy choice — matters more than algorithm choice in this experimental setting.

TABLE VI. Detailed performance metrics of individual optimization runs, categorized by algorithm, strategy, and initialization mode.

Algo	Strategy	Run ID	Start Mode	Train Final Cash	Test Final Cash
mrfo	ema_shared	19	random	94,671.432702	11,973.202678
pso	ema_shared	15	fixed	103,623.884175	10,074.639102
sos	ema_shared	14	random	99,739.498022	7,860.381078
aos	ema_shared	25	random	53,162.066536	7,860.381078
abo	ema_shared	6	random	61,669.572182	7,795.072801
aos	ema_independent	3	fixed	8,387.916226	7,478.455514
sos	ema_shared	22	fixed	104,921.108049	7,142.090134
aos	ema_independent	1	random	7,387.991921	7,041.993653
sos	ema_shared	4	fixed	85,227.061950	6,859.085979
sos	ema_shared	3	random	120,633.526793	6,661.169413
aos	ema_shared	11	random	34,442.798674	6,144.024916
abo	ema_shared	6	fixed	82,411.179264	6,138.875681
pso	ema_shared	0	fixed	79,852.603444	6,128.390758
pso	weighted	28	random	13,408.479220	5,965.439091
abo	ema_shared	14	random	76,429.269569	5,906.325814
abo	weighted	9	random	10,015.651104	5,853.488490
abo	weighted	24	random	10,757.072994	5,767.026518
aos	ema_independent	28	fixed	6,960.173097	5,744.356481
aos	weighted	6	random	11,633.609194	5,737.972241
abo	weighted	8	fixed	11,447.583852	5,691.006969
gps	ema_independent	4	fixed	7,365.586603	5,684.316408
mrfo	weighted	20	random	11,545.351863	5,680.524618
abo	weighted	23	random	10,582.572136	5,677.242799
aos	ema_independent	7	fixed	6,152.365887	5,675.507260
mrfo	weighted	14	random	11,877.068017	5,670.184694
mrfo	ema_shared	14	fixed	79,302.563503	5,660.263187
mrfo	ema_shared	1	random	49,408.611819	5,660.263187

F. Interpretation

Two non-exclusive explanations account for the failure of training fitness to generalise. The first is overfitting to the training market regime: parameters optimised on the pre-2020 BTC market exploit structural patterns — momentum cycles, volatility regimes — specific to that period. The COVID-19 crash, the 2021 bull run, and the 2022 bear market introduce qualitatively different dynamics that these parameters are not positioned to exploit.

The second explanation is that MA crossover signals are fundamentally weak predictors of BTC price movement across diverse regimes regardless of how precisely the parameters are tuned; if strategy expressiveness is the binding constraint, no optimiser can recover transferable advantage from the parameter space. The 1.44% rate of bots exceeding buy-and-hold across 1,800 trials is statistically consistent with random sampling from the strategy-fitness right tail, and does not provide evidence of a learnable trading edge under the MA-crossover hypothesis space. This is consistent with — though does not formally test — the efficient market hypothesis for short-horizon technical signals on highly liquid assets.

VI. CONCLUSION

Algorithm choice has a universally significant effect on training fitness across all strategy and initialisation conditions, yet this advantage does not transfer to test performance. The binding constraint appears to be strategy expressiveness in the face of market non-stationarity rather than optimiser quality: once the target regime shifts, the parameters optimised on training data no longer generalise regardless of how effectively they were found.

AOS performs comparably to PSO despite lacking personal best memory. On this multimodal fitness landscape, global best guidance via nucleus tracking combined with shell-based population distribution appears sufficient; the additional per-particle history that PSO maintains provides no statistically significant advantage. This is a finding specific to the problem structure — the MA crossover landscape may not present the kind of elongated, directional ridges where personal memory typically helps.

GPS confirms that population diversity is necessary. Single-state local search is consistently and significantly worse



FIG. 4. Dunn's post-hoc pairwise p -values (Bonferroni-corrected) for all strategy $\times$ initialisation combinations — test fitness.

than every population-based method across all conditions. The MA parameter landscape contains enough local optima that a deterministic pattern search from a fixed starting point cannot reliably reach high-fitness regions, establishing population-based global search as a structural requirement rather than an optional improvement.

SOS's three-evaluation-per-iteration cost is not justified by its performance differential over PSO. Because Dunn's test establishes distributional equivalence between PSO and SOS on training fitness under matched-iteration budgets, the threefold evaluation overhead buys no statistical advantage. A compute-normalised comparison equalising total fitness evaluations rather than iterations would be required to confirm this quantitatively, but the direction of the finding is clear.

Independent α in `ema_independent` provides genuinely greater expressiveness than the shared- α `ema_shared` configuration. The expanded search space produces more pronounced inter-algorithm differences (higher Kruskal-Wallis H -statistics), indicating that the additional degrees of freedom challenge the optimisers' exploration capabilities in a way that better reveals their structural differences.

The LE degradation in AOS is an empirically observed phenomenon. The lowest-energy electron's role as a local-

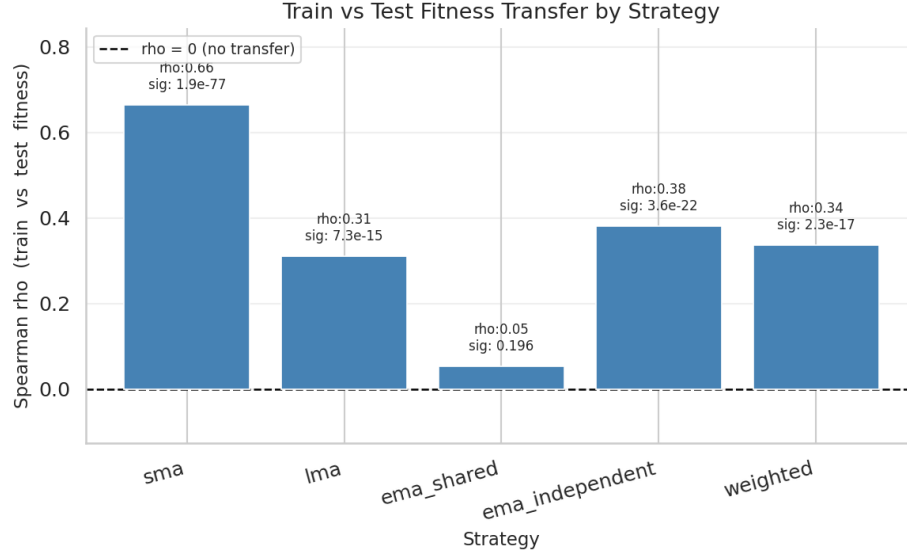


FIG. 5. Spearman rank correlation between training and test fitness per strategy (both initialisation modes pooled, $n = 360$ per strategy). $\rho = 0$ corresponds to no rank transfer.

optima escape mechanism degrades as the nucleus shifts across iterations and other electrons improve, because the LE designation tracks relative ranking within the current population rather than an absolute landscape property. This is a genuine weakness in the paper’s formulation that warrants further investigation.

The no-trade degenerate optimum is an unpenalised artifact of using raw final cash as the fitness metric. A risk-adjusted metric such as the Sharpe ratio or return-per-trade would penalise conservative non-trading behaviour and remove this fixed point from the feasible optima.

Outlook Future work should adopt risk-adjusted fitness functions and regime-aware train/test splits that better reflect the non-stationarity of cryptocurrency markets. The train/test divergence observed here suggests the evaluation objective itself may require redesign: objectives that penalise in-sample overfitting — for example, through cross-validated segment fitness or walk-forward testing — would impose generalisation pressure that the current segment-average fitness does not provide.

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TABLE VII. Updated statistical summary of performance metrics sorted by strategy configuration and coefficient of variation (CV) under the **fixed** start mode across 30 runs per cell.

Algo	Strategy	Count	Mean	Std	CV	Var	Min	Median	Max
mrfo	ema_independent	30	1,000.0000	0.0000	0.0000	0.0000	1,000.0000	1,000.0000	1,000.0000
aos	ema_independent	30	4,271.8511	1,401.7027	0.3281	1,964,770.5024	977.9262	4,425.3908	7,478.4555
sos	ema_independent	30	3,805.2664	1,285.2927	0.3378	1,651,977.3254	1,206.8093	4,296.9697	4,834.2114
pso	ema_independent	30	3,219.7871	1,460.6491	0.4536	2,133,495.7079	1,201.2297	4,296.9697	4,565.8008
abo	ema_independent	30	1,482.4498	994.6911	0.6710	989,410.4156	858.1458	1,202.4195	4,360.9038
gps	ema_independent	30	2,328.6670	1,619.2726	0.6954	2,622,043.6733	509.7981	1,588.7568	5,684.3164
aos	ema_shared	30	1,894.0325	1,269.0238	0.6700	1,610,421.4700	603.2835	1,137.1210	4,377.1504
sos	ema_shared	30	2,237.5683	1,698.3336	0.7590	2,884,337.1312	617.0044	1,862.0257	7,142.0901
abo	ema_shared	30	1,815.8120	1,391.7397	0.7665	1,936,939.2930	633.3265	1,157.6009	6,138.8757
gps	ema_shared	30	1,831.8888	1,465.6274	0.8001	2,148,063.7275	462.4916	1,000.0000	4,584.5951
mrfo	ema_shared	30	1,606.1015	1,366.8988	0.8511	1,868,412.3809	484.1837	1,000.0000	5,660.2632
pso	ema_shared	30	2,468.1415	2,176.0318	0.8816	4,735,114.2306	824.6249	1,000.0000	10,074.6391
sos	lma	30	907.9777	0.0000	0.0000	0.0000	907.9777	907.9777	907.9777
aos	lma	30	794.0966	176.2895	0.2220	31,077.9891	485.4056	907.9777	921.2818
pso	lma	30	587.2506	135.8814	0.2314	18,463.7669	487.7403	487.7403	907.9777
gps	lma	30	879.1397	221.1375	0.2515	48,901.8002	378.9697	942.9760	1,228.5388
mrfo	lma	30	749.9327	200.3511	0.2672	40,140.5782	487.7403	907.9777	907.9777
abo	lma	30	683.1384	187.4051	0.2743	35,120.6796	487.7403	671.5093	907.9777
mrfo	sma	30	1,180.2757	0.0000	0.0000	0.0000	1,180.2757	1,180.2757	1,180.2757
sos	sma	30	1,180.2757	0.0000	0.0000	0.0000	1,180.2757	1,180.2757	1,180.2757
abo	sma	30	1,172.2460	12.4752	0.0106	155.6317	1,153.5101	1,180.2757	1,180.2757
aos	sma	30	1,172.2460	12.4752	0.0106	155.6317	1,153.5101	1,180.2757	1,180.2757
pso	sma	30	1,129.0634	142.7064	0.1264	20,365.1027	617.3679	1,180.2757	1,180.2757
gps	sma	30	1,154.9955	216.3662	0.1873	46,814.3537	725.7645	1,104.1544	1,740.1289
pso	weighted	30	4,813.3528	460.0434	0.0956	211,639.9732	3,637.1189	4,729.4264	5,564.7885
sos	weighted	30	4,580.5727	477.9724	0.1043	228,457.5786	2,050.8875	4,660.9005	4,694.2637
abo	weighted	30	4,451.6966	817.0929	0.1835	667,640.8260	1,130.4039	4,651.6147	5,691.0070
aos	weighted	30	4,465.4214	830.2398	0.1859	689,298.1441	1,000.0000	4,676.3205	5,500.0430
mrfo	weighted	30	4,329.0829	1,145.6547	0.2646	1,312,524.6257	1,381.9554	4,694.2197	5,644.1051
gps	weighted	30	2,904.4636	1,852.4269	0.6378	3,431,485.2697	370.9620	3,688.6018	5,093.2604